



Micro-Jet Engine for UAV applications

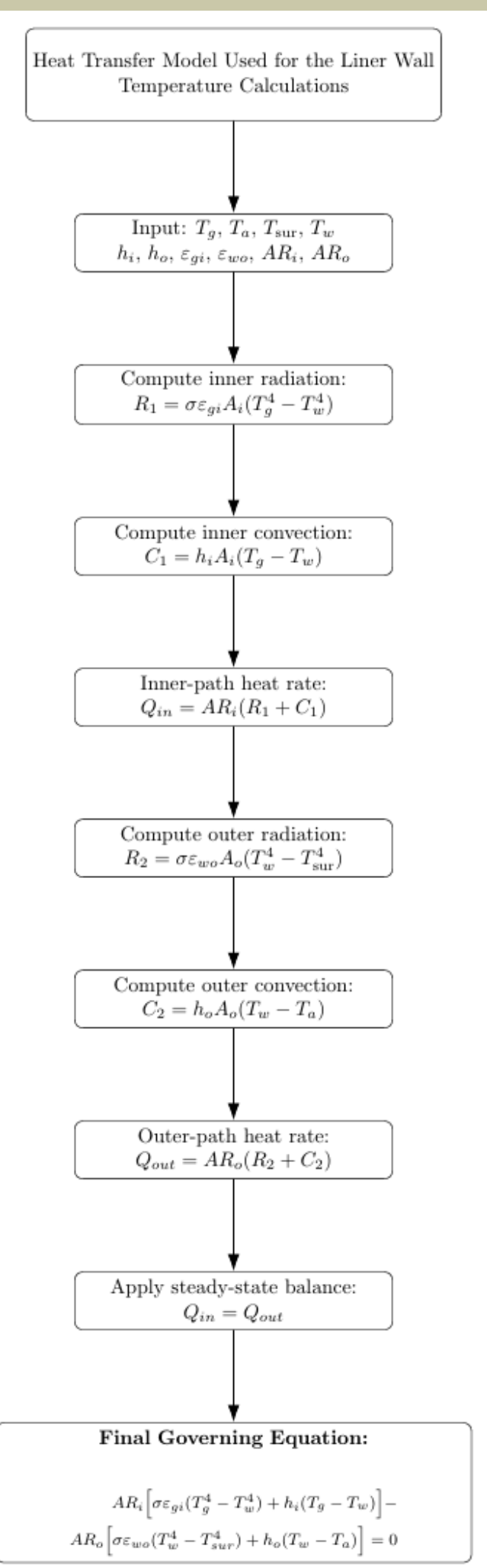
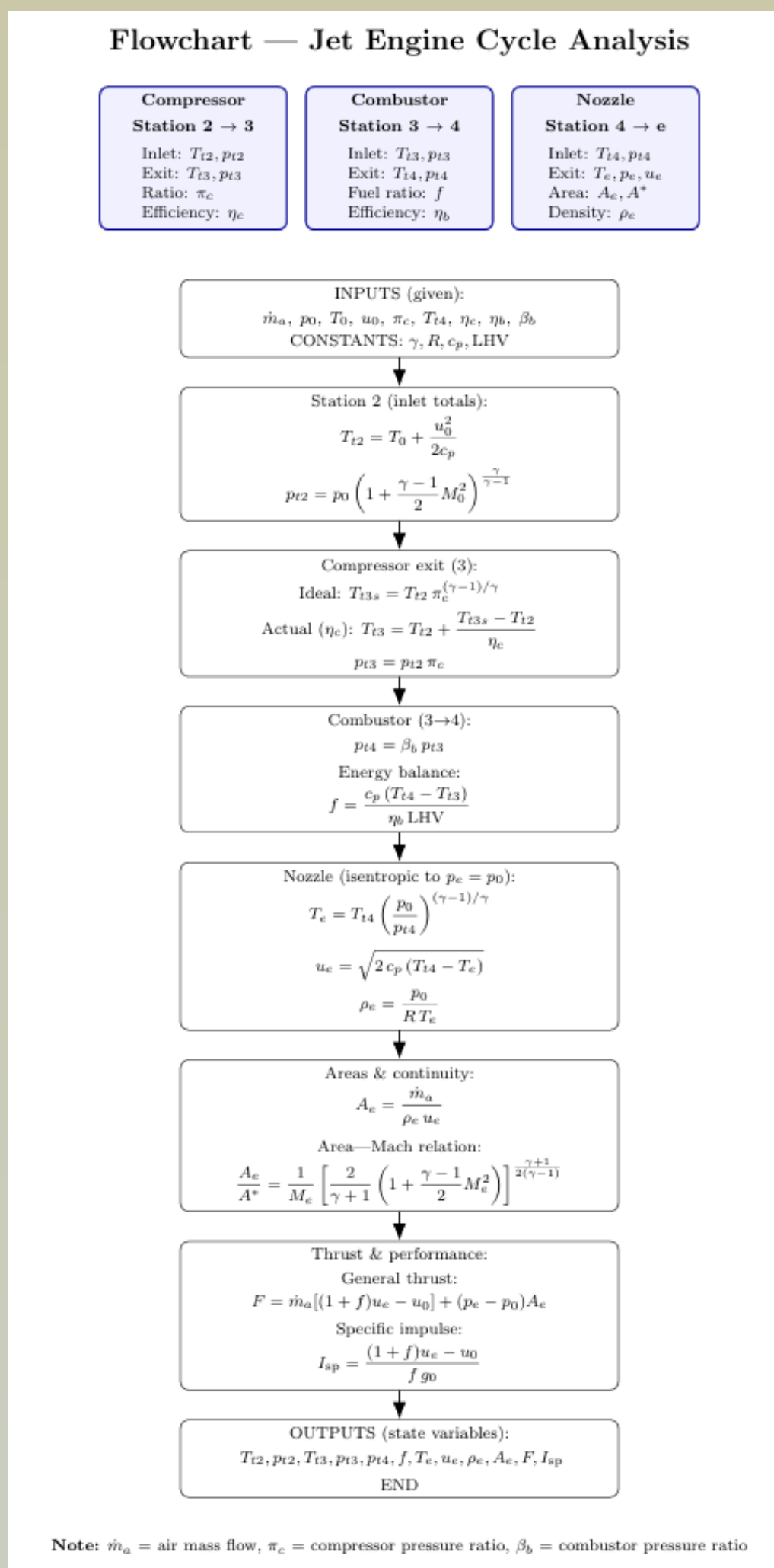
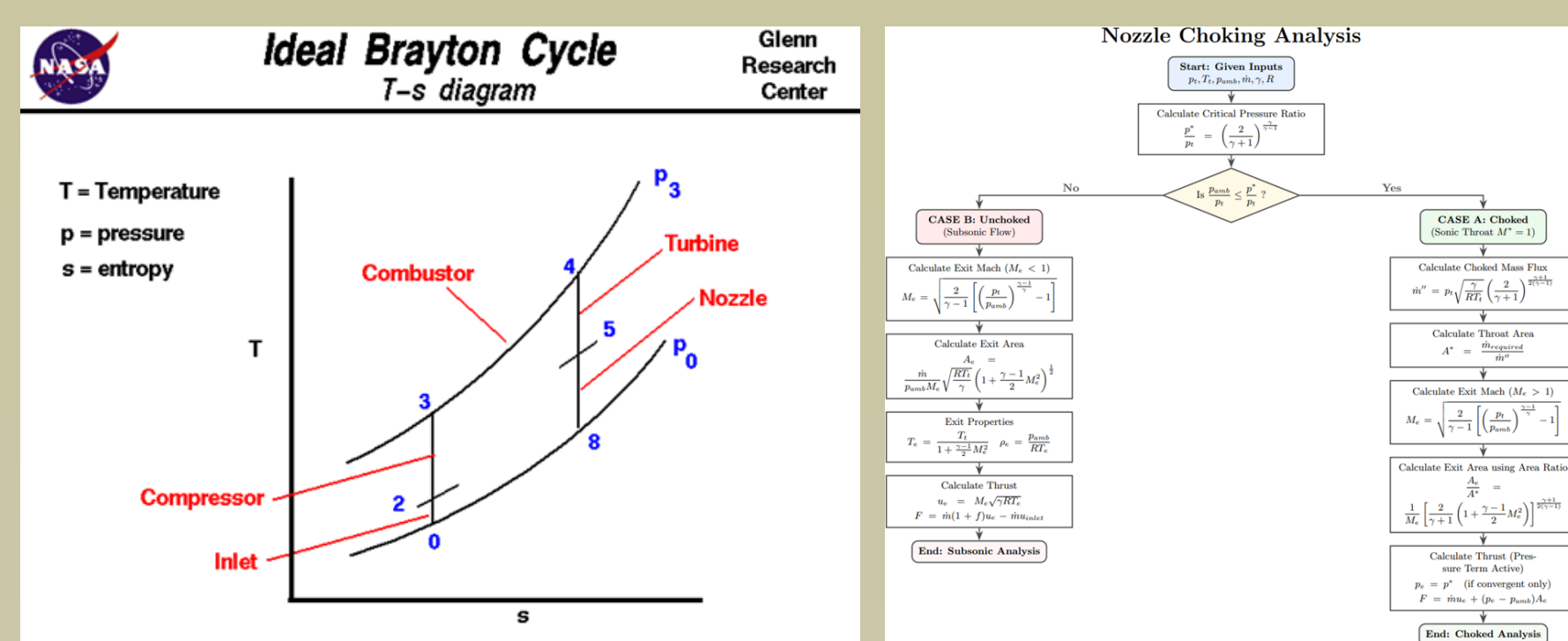
CP-302 Project under the supervision of Dr.Rajendra Munian & Dr. Srikant Padhee

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Introduction

- Micro-jet engines are increasingly used in UAVs due to their compact size, high power-to-weight ratio, and ability to deliver sustained thrust for long-duration missions.
- This project focuses on developing a thrust-controllable and fully functional micro-jet engine designed specifically for UAV applications.
- Key planned performance targets include a maximum thrust of ~30 kg, an overall diameter under 15 cm, and the ability to run reliably on petrol fuel, which is more accessible and cost-effective than kerosene-based alternatives.
- Core components follow the Brayton cycle layout: air intake, centrifugal compressor, combustion chamber, turbine, and convergent-divergent nozzle.
- Computational tools such as Ansys CCD design, Blade Gen, Turbogrid(For Designing the impeller), and Ansys Fluent (For estimationg the pressure increase across the diverging duct).
- A locally hosted ESP32 web server enables seamless, low-latency control of the micro-jet engine through any browser on the network. Using DHCP and optional mDNS, the system provides easy IP access, while a WebSocket channel ensures real-time, bidirectional communication without page reloads. User commands sent as JSON are instantly processed and translated into PWM outputs, enabling precise and responsive throttle and subsystem control.

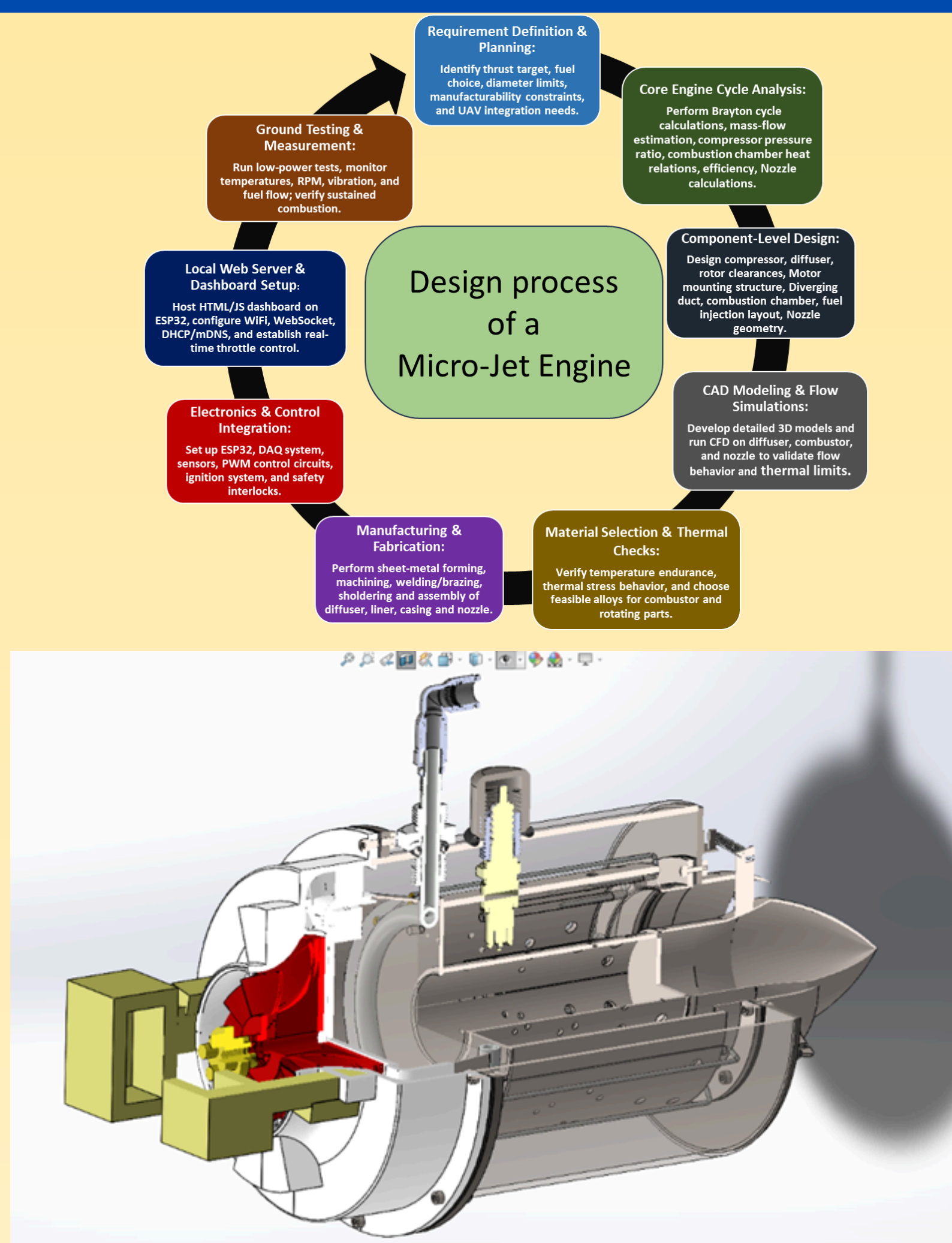
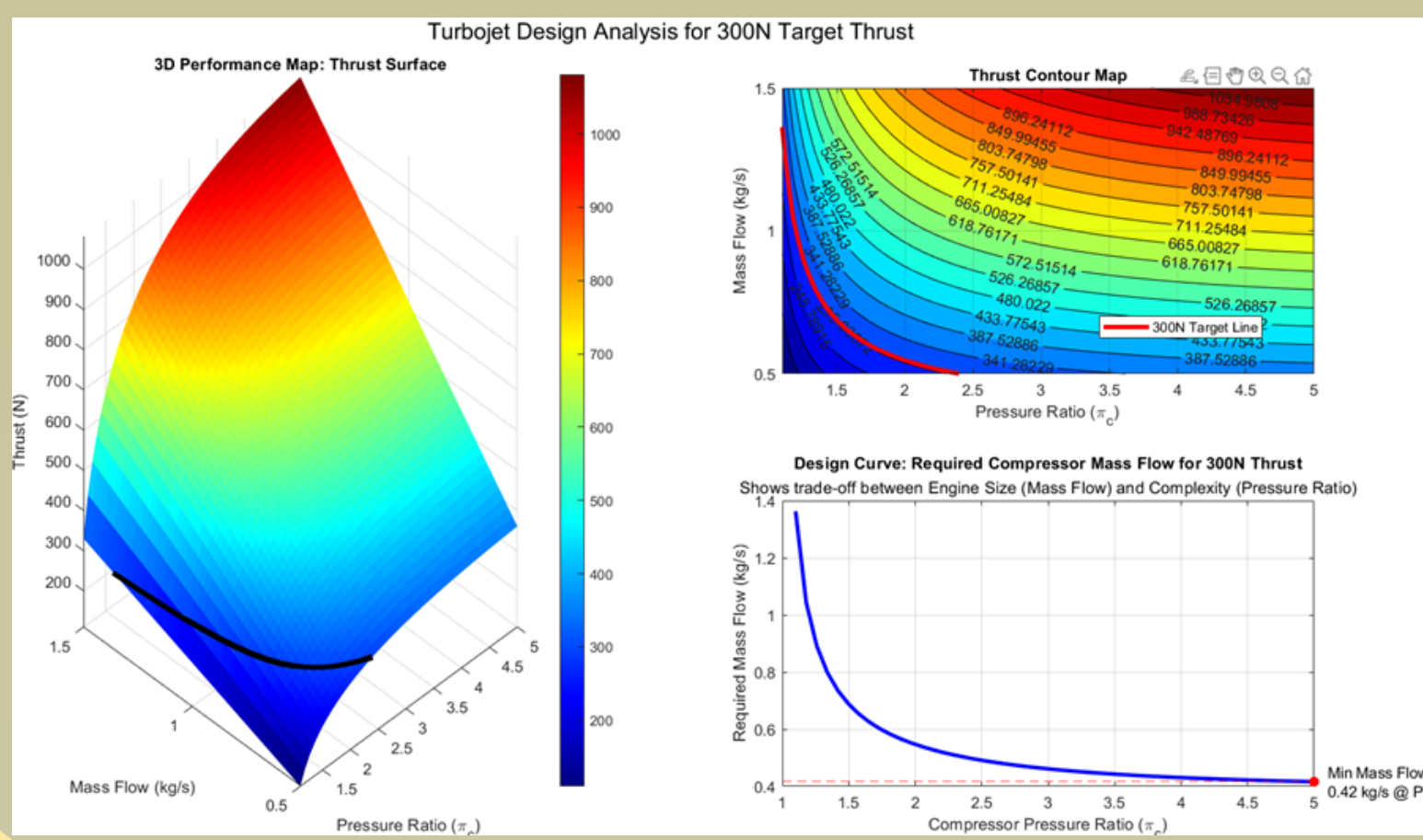
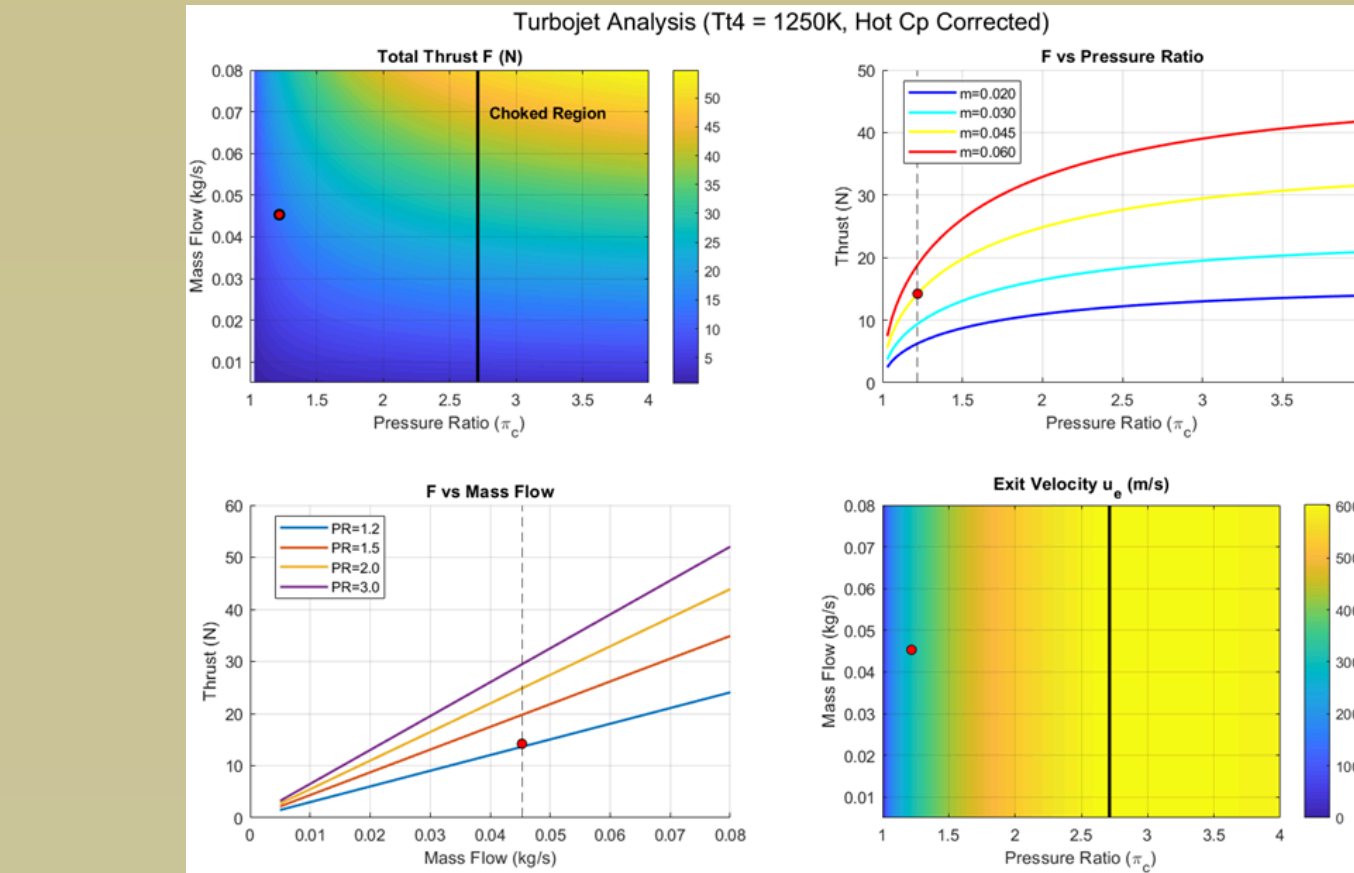
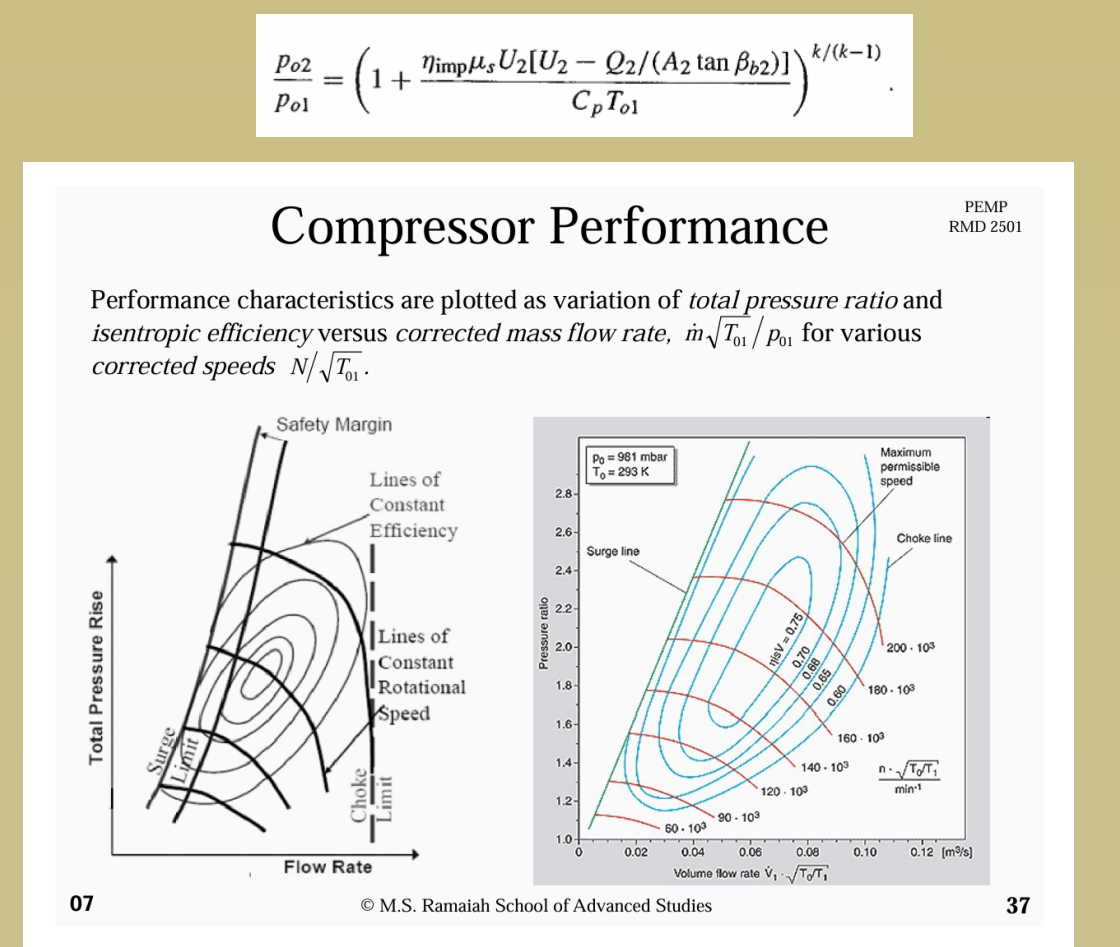
Thermodynamic Engine Analysis



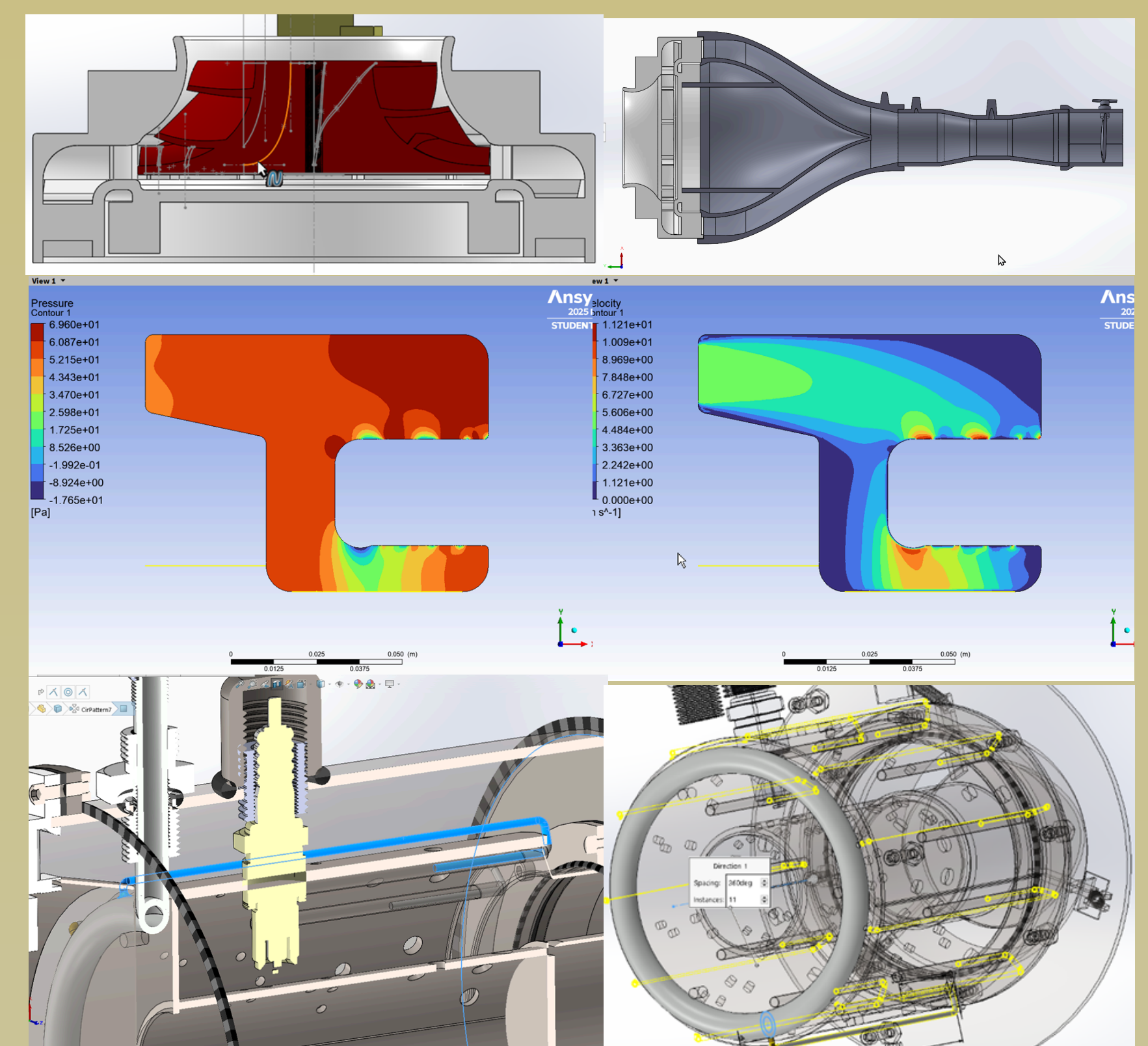
Variable	Description	Value / Formula	Unit
p0	Static Ambient Pressure	100000	Pa
V2	Inlet Velocity	2.86	m/s
T0	Static Ambient Temperature	308.526	K
rho2	Static Density	1.13	kg/m³
m_dot_a	Air Mass Flow Rate	0.0453	kg/s
gamma	Ratio of Specific Heats	1.4	-
R	Gas Constant (Air)	287	J/(kg·K)
cp	Specific Heat	1005	J/(kg·K)
LHV	Lower Heating Value	43000000	J/kg
eta_b	Combustor Efficiency	0.85	-
T4	Turbine Inlet Temperature	1230	K
PR	Compressor Pressure Ratio	1.22	-
eta_c	Compressor Efficiency	0.8	-
a2	Speed of Sound at Inlet	340	m/s
M2	Mach Number	0.008736882	-
T3	Comp. Exit Temp (Actual)	331.071372	K
p3	Comp. Exit Pressure	122000	Pa
p4	Stagnation Pressure (Comb. Exit)	122000	Pa
f	Fuel-to-Air Ratio	0.033614979	-
Te	Nozzle Exit Static Temp	1185.951598	K
ue	Exit Velocity	372.5146814	m/s
u0	Flight Speed (Static Test)	0	m/s
Spec. Thrust	Specific Thrust	384.8233036	N/(kg/s)
T4	Turbine Inlet Temperature	1230	K
m_dot_f	Fuel Mass Flow Rate	0.001496802	kg/s
Isp	Specific Impulse	26207.65159	s

State variables estimation at different Stages.

Component level design



CAD modelling & Flow simulations



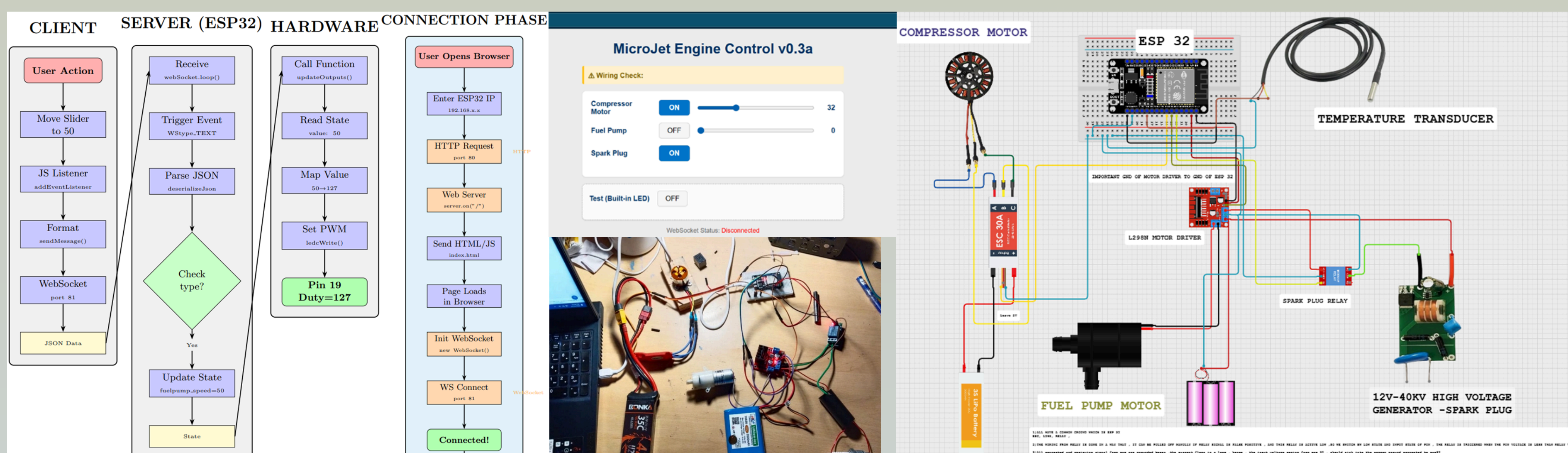
1.2MM ID Injection Tubes (Blue), passing over the outer liner, carries fuel which absorbs heat from the outer liner evaporating the fuel before injection for better mixing.

Manufacturing & Fabrication



5' outer liner, 1 mm MS sheet, rubber gaskets; laser cutting of MS sheet and finishing operations; drilling on 15.6 cm MS tube outer liner; arc-welding of inner liner, outer liner, and casing; 20-nut joint assembly between combustion chamber and casing; brazing of evaporator and fuel-injection tubes; spark plug installation and fuel-nozzle forming/welding.

Electronics & Web based control integration



This ESP32-based electronics and web control system is used for operating the micro jet engine. The ESP32 receives commands from a web interface and controls the compressor motor speed, fuel pump for regulated fuel supply, and triggers the spark plug via PWM. The integrated setup allows instantaneous real-time remote control and data collection from all engine components.